

Sheet 3: Instruments

1. A pressure gauge with a range between 0 - 10 bar is found to have an error of ± 0.15 bar when calibrated by the manufacturer.

Calculate:

- The error percentage of the gauge.
 - The error percentage when the reading obtained is 2.0 bar.
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Solution:

a.
$$\text{Error Percentage} = \pm \frac{0.15 \text{ bar}}{10 \text{ bar}} \times 100 = \pm 1.5 \%$$

b.
$$\text{Error Percentage} = \pm \frac{0.15 \text{ bar}}{2 \text{ bar}} \times 100 = \pm 7.5 \%$$

The gauge is not suitable for use for low range reading.

Alternative : use gauge with a suitable range.

2. Two pressure gauges (pressure gauge A and B) have a full scale accuracy of $\pm 5\%$. Sensor A has a range of 0-1 bar and Sensor B 0-10 bar. Which gauge is more suitable to be used if the reading is 0.9 bar?

Solution:

Sensor A :

$$\text{Equipment max error} = \pm \frac{5}{100} \times 1 \text{ bar} = \pm 0.05 \text{ bar}$$

Equipment accuracy

$$0.9 \text{ bar (in \%)} = \pm \frac{0.05 \text{ bar}}{0.9 \text{ bar}} \times 100 = \pm 5.6 \%$$

Sensor B :

$$\text{Equipment max error} = \pm \frac{5}{100} \times 10 \text{ bar} = \pm 0.5 \text{ bar}$$

Equipment accuracy

$$0.9 \text{ bar (in \%)} = \pm \frac{0.5 \text{ bar}}{0.9 \text{ bar}} \times 100 = \pm 55 \%$$

Conclusion :

Sensor A is more suitable to use at a reading of 0.9 bar because the error percentage ($\pm 5.6\%$) is smaller compared to the percentage error of Sensor B ($\pm 55\%$).

3. A pressure gauge of range 0-20 bar is said to have an error of ± 0.25 bar when calibrated by the manufacturer. Calculate the percentage error on the basis of maximum scale value. What would be the possible error as a percentage of the indicated value when a reading of 5 bar is obtained in a test?

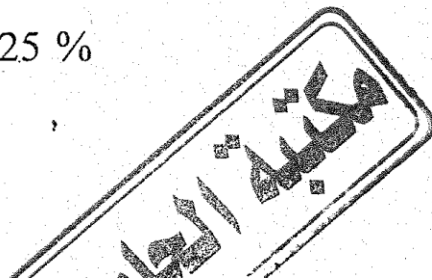
Solution:

$$\text{Pressure gauge max error} = \pm \frac{\text{Percentage error (accuracy)}}{100} \times 20 \text{ bar} = \pm 0.25 \text{ bar}$$

$$\text{Percentage error (accuracy)} = \pm \frac{0.25}{20} \times 100 = 1.25 \%$$

Pressure gauge accuracy

$$5 \text{ bar (in \%)} = \pm \frac{0.25 \text{ bar}}{5 \text{ bar}} \times 100 = \pm 5 \%$$



4. A spring scale requires a change of 15kg_f in the applied weight to produce a 2 cm change in the deflection of the spring scale. Determine the static sensitivity.

Solution:

SENSITIVITY



- Defined as the ratio of change in output towards the change in input at a steady state condition.

$$\text{Sensitivity (k)} = \frac{\Delta \theta_o}{\Delta \theta_i}$$

$\Delta \theta_o$: change in output; $\Delta \theta_i$: change in input

$$\text{Sensitivity (k)} = \frac{\text{Scale deflection}}{\text{Value of measurand producing deflection}}$$

$$\text{Sensitivity (k)} = \frac{2 \text{ cm}}{15 \text{ kg}} = 1.333 \text{ mm / kg}$$

5. Explain the following statements:

- (i) A galvanometer has a sensitivity specified of 15 mm/ μ A.
 - (ii) A Platinum Resistance Thermometer has a sensitivity specified of 0.5 $\Omega/^{\circ}\text{C}$
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Solution:

- (i) If the change in the current (input) equals 1 μ A, the change in the output (deflection of the instrument) will be 15 mm at the steady state.
- (ii) If the change in the temperature (input) equals 1 $^{\circ}\text{C}$, the change in the output (resistance) will be 0.5 Ω at the steady state.

6. The output of a platinum resistance thermometer (RTD) is as follows:

Input($^{\circ}\text{C}$)	Output(Ohm)
0	0
100	200
200	400
300	600
400	800

Calculate the sensitivity of the equipment.

Solution:

Draw an input versus output graph. From that graph, the sensitivity is the slope of the graph.

$$\text{Sensitivity (k)} = \frac{\Delta \theta_o}{\Delta \theta_i} = \frac{(400 - 200) \text{ ohm}}{(200 - 100)^{\circ}\text{C}} = 2 \text{ ohm}/^{\circ}\text{C}$$

7. A measuring system consists of a transducer, an amplifier and a recorder, and their individual sensitivities are stated as follows:

Transfer sensitivity $K_1 = 0.25 \text{ mV}/^{\circ}\text{C}$

Amplifier gain $K_2 = 2.5 \text{ V/mV}$

Recorder sensitivity $K_3 = 4 \text{ mm/V}$

What would be the overall sensitivity of the measuring system?

Solution:

Sensitivity of the whole system:

$$k = k_1 \times k_2 \times k_3 = \frac{0.25 \text{ mV}}{^{\circ}\text{C}} \times \frac{2.5 \text{ V}}{\text{mV}} \times \frac{4 \text{ mm}}{\text{V}} = 2.5 \text{ mm}/^{\circ}\text{C}$$

8. A pressure measuring system consists of a piezoelectric transducer, a charge amplifier and a ultra violet charge recorder. The sensitivities of these elements are stated as follows :

Piezoelectric transducer, $K_1 = 8.5 \text{ PC/bar}$

Charge amplifier, $K_2 = 0.004 \text{ V/PC}$

Ultraviolet charge recorder, $K_3 = 20 \text{ mm/V}$

What would be the deflection on the chart due to a pressure change of 30 bar?

Solution:

Sensitivity of the whole system:

$$k = k_1 \times k_2 \times k_3 = \frac{8.5 \text{ pC}}{\text{bar}} \times \frac{0.004 \text{ V}}{\text{pC}} \times \frac{20 \text{ mm}}{\text{V}} = 0.68 \text{ mm / bar}$$

$$\therefore \text{Sensitivity (k)} = \frac{\text{Scale deflection (mm)}}{\text{Value of measurand producing deflection (bar)}}$$

$$\therefore \text{deflection} = 0.68 \times 30 = 20.4 \text{ mm}$$

9. (a) An instrument is calibrated in an environment at a temperature of 20°C and the following output readings y are obtained for various input values x:

y	13.1	26.2	39.3	52.4	65.5	78.6
x	5	10	15	20	25	30

Determine the measurement sensitivity, expressed as the ratio y/x.

- (b) When the instrument is subsequently used in an environment at a temperature of 50°C, the input/output characteristic changes to the following:

y	14.7	29.4	44.1	58.8	73.5	88.2
x	5	10	15	20	25	30

Determine the new measurement sensitivity. Hence determine the sensitivity drift due to the change in ambient temperature of 30°C.

Solution:

At 20°C
$$\text{Sensitivity} = \frac{\Delta y}{\Delta x} = \frac{26.2 - 13.1}{10 - 5} = 2.62$$

At 50°C
$$\text{Sensitivity} = \frac{\Delta y}{\Delta x} = \frac{29.4 - 14.7}{10 - 5} = 2.94$$

$$\text{Sensitivity drift} = \text{Sensitivity}_{50} - \text{Sensitivity}_{20} = 2.94 - 2.62 = 0.32$$

10. A load cell is calibrated in an environment at a temperature of 21°C and has the following deflection/load characteristic:

Load (kg)	0	50	100	150	200
Deflection (mm)	0	1	2	3	4

When used in an environment at 35°C, its characteristic changes to the following:

Load (kg)	0	50	100	150	200
Deflection (mm)	0.2	1.3	2.4	3.5	4.6

- (a) Determine the sensitivity at 21°C and 35°C.
 (b) Calculate the total zero drift and sensitivity drift at 35°C.
 (c) Hence determine the zero drift and sensitivity drift coefficients (in units of $\mu\text{m}/^\circ\text{C}$ and $(\mu\text{m per kg})/^\circ\text{C}$).
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Solution:

a. At 21°C Sensitivity = $\frac{\Delta y}{\Delta x} = \frac{1-0}{50-0} = 20 \mu\text{m/kg}$

At 35°C Sensitivity = $\frac{\Delta y}{\Delta x} = \frac{1.3-0.2}{50-0} = 22 \mu\text{m/kg}$

b. Zero drift = $0.2 \text{ mm} = 200 \mu\text{m}$

Sensitivity drift = Sensitivity₃₅ - Sensitivity₂₁ = $22 - 20 = 2 \mu\text{m/kg}$

c. Zero drift/ $^\circ\text{C} = \frac{200}{14} = 14.2857 \mu\text{m}/^\circ\text{C}$

Sensitivity drift/ $^\circ\text{C} = \frac{2}{14} = 0.1429 (\mu\text{m per kg})/^\circ\text{C}$